Class Meeting Handout #2

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This document contains information for the discussion and activities of our second Chemistry 4410 class meeting.

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Diagrams were created using *Affinity Designer* and chemical structure were made using *ChemDoodle*.

Molecular Orbitals

Dots, lines and arrows are often used to describe lone pairs, covalent bonds and electron movement in reactions. These are representations of a deeper truth: molecular orbitals. You will already have reviewed this topic by reading chapters 1.1 and 1.2 as instructed on the Moodle site.[1] Now let's practice.

[1] Completing the assigned readings and/or problems listed in the *Before* section of the Moodle page for this class meeting will help you participate more fully in class exercises.

We will be using a model for molecular orbitals that uses the sp, sp^2 and sp^3 hybrid orbitals of atoms as the starting point for building local molecular orbitals that describe bonds and groups. We don't need to consider the entire molecule to explain most effects that are observed as structural changes alter reactivity. Today, we will practice this simplified model.

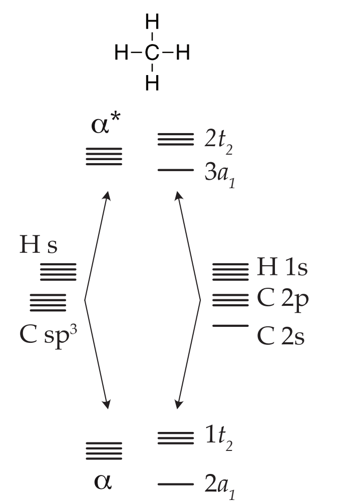

Figure 1: Getting started with methane.↑ Simplified local MOs constructed from hybrid orbitals are compared to the more accurate MOs constructed from atomic orbitals using symmetry rules and trigonometry. Compare to figure 40 in the "Molecular Orbital Tutorial" available via the Moodle page for this class meeting.

Review: Build a Basis Set – 10 minutes

Consider methane, ethane, ethylene, and formaldehyde.

1. Draw the Lewis structure for each of the molecules.

2. Identify polar bonds, partial charges and dipoles. Comment on the expected reactivity.

3. Draw an energy diagram for all the atomic orbitals. Use hybrid atomic orbitals that match the geometry of each atom. See Figure 1 for an example with methane.

Exercise: Combine Basis Set Orbitals – 20 minutes

Using the basis sets of atomic orbitals established above…

4. Combine all atomic orbitals involved in bonding interactions to create the bonding and antibonding molecular orbitals for each bond. Place these molecular orbitals on the energy diagram.[2,3]

5. Identify all lone pairs and the atomic orbital for which they are associated. Place these unused non-bonding molecular orbitals on the energy diagram.

6. Identify the frontier molecular orbitals that best explain the reactivity of the molecules.

[2] The textbook presents a more rigorous method based on Walsh orbitals. It is better and more accurate. However, I believe that the simplified approach used in this class meeting is all that we need. Follow along with me and compare our systems with those presented in chapter 1.2 of the textbook.

[3] When I am not around, consult the "Molecular Orbital Tutorial" available on the Moodle page for this class meeting. It's my personal manifesto on combining atomic orbitals to construct local molecular orbitals.

Orbitals are the Truth

Keeping local molecular orbitals in mind will help us to propose more accurate mechanisms.

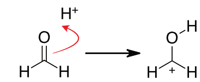

Figure 2: Protonation of formaldehyde

Review: Deeply Conscious Arrow Pushing – 10 minutes

A mechanism for the protonation of formaldehyde is shown in figure 2. Using the molecular orbitals of the molecules involved…

7. Evaluate the proposed mechanism.[4]
8. Propose a mechanism in which the arrows better represent the molecular orbital changes.

[4] It's not correct, but it's also not wrong. It is a commonly used and also misleading shortcut. In this course, we will purify our minds and put such heresies behind us. Our mechanisms will use the truth of frontier molecular orbitals as a foundation.

Exercise: The Anomeric Effect – 10 minutes

Consider the *axial* and *equatorial* conformers of 2-methoxytetrahydropyran as shown in figure 3…

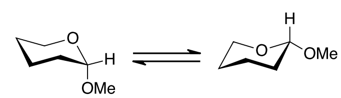

Figure 3: Conformers of 2-methoxytetra-hydropyran.

9. Using the data from table 2.14 of the textbook[5], propose which is more stable and estimate the equilibrium constant.
10. The equilibrium population is observed to favour the *axial* conformer with an equilibrium constant of approximately $K_{eq} = 5$. Explain this observation.[6,7]

[5] See page 104 of the textbook and recall the activity from the previous class meeting.

$$K_{eq} = e^{-\frac{\Delta G^\circ}{RT}}$$
$$R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$$

[6] Hint: we have just spent 40 minutes exploring molecular orbitals and their interactions.

[7] Today wasn't all review. You will have already encountered the new idea of the *anomeric effect* as you read chapter 2.4 of the textbook, specifically in section 2.4.2

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing all additional requirements,[8] you will have achieved the following…

1. Able to draw a molecular orbital diagram for a simple molecule or for a functional group in a larger molecule and use it to interpret reactivity.
2. Been introduced to how electronic effects can be as important as steric effects in interpreting how structure affects energy.

[8] Be sure to attempt the textbook problems suggested on the Moodle page for this class meeting. These problems will apply what we have learned in this meeting and explore the concepts covered in the textbook that we did not have time to explore together. Fifty minutes is never enough; fortunately, you have the textbook to help you fill in the gaps.

Looking Ahead

We will be using the careful arrow pushing that we introduced today to review some basic organic reaction mechanisms over the next few meetings. We also have been introduced to how the electronic effect of changing a carbon atom to an oxygen in methoxycyclohexane can change the outcome of the interconversion between conformers. Later in the course, we will examine the electronic effect of substituents in great detail.